

Problem Statement

The sonic-mgmt test framework is a critical component of PR validation and continuous integration. However, over time, several inefficiencies have accumulated:

- Some individual test scripts run longer than 30 minutes, and some individual test cases exceed 10 minutes, PR pipelines retry failed test scripts up to two additional times, amplifying runtime cost.
- Long-running scripts disproportionately consume KVM CPU resources.
- Excessive runtime increases PR turnaround time and reduces CI throughput.

Currently, there is no formal runtime governance framework, no enforcement threshold, and limited visibility into runtime regressions across scripts, test cases, topology types, and branches.

As SONiC continues to scale, unmanaged test runtime growth will:

- Increase infrastructure cost
- Slow down PR validation
- Reduce contributor productivity
- Impact overall CI system stability

This project aims to establish a data-driven runtime governance and optimization framework for sonic-mgmt.

Mentorship Objectives

The mentee(s) will work on designing and implementing a structured runtime optimization initiative that includes:

- Refactor selected high-impact long-running test scripts
- Propose test case optimization strategies:
 - Reduce loop/iteration counts
 - Reduce coverage where representative sampling is sufficient
 - Optimize setup/teardown overhead
 - Remove redundant waits

Expected Outcomes and Deliverables

By the end of the mentorship program, the following deliverables are expected:

1. Refactored high-impact test scripts demonstrating measurable improvement
2. A structured runtime dataset and analysis report
3. A documented SONiC Test Runtime Best Practices guide

Recommended Qualifications

- Python programming skills
- Familiarity with pytest framework
- Basic understanding of CI/CD systems
- Ability to analyze data (Kusto / SQL / Pandas experience preferred)
- Understanding of networking fundamentals (basic SONiC knowledge preferred)

Mentor(s) Names and Contact Info

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Repo: <https://github.com/sonic-net/sonic-mgmt>

Region/Term: China and Summer